Data & Decisions: Section 2

Confidence Intervals and Hypothesis Testing

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Next Deadlines

- Knowledge Check 3 - due Thu 10/22
- Team Assignment - due Thu 10/22
- Usual weekly OHs (Reed's or Silvia's) for YOUR doubts
- See you in Berkeley for Block 3

Questions?

Roadmap

1. Review Ch. 15

Confidence Intervals

2. Review Ch. 16

Statistical Tests

3. Review Ch. 17

Experimental Setting

Review Ch. 15

CLs vs CIs

- Both rely on the **sampling distribution** of a statistic (NO paramater!)
- **Control limits**
  - **We know the population parameters**
  - They determine the range in which $\bar{X}$ should be found
  - We extract a sample and we have a rule to determine whether our process is working properly (i.e. the sample is extracted from the right population)
- **Confidence Intervals**
  - **We don't know the population parameters**
  - We extract a sample and we have a range that should hold the population parameter

Control Limits

- Already discussed during Block 1
- Please look at the exercise uploaded in bCourses
  - Need to know which problem you are dealing with and how to solve it

Confidence Intervals

- **Confidence Intervals** provides a range of **plausible values** for an **unknown parameter** based on your sample
- The length of the interval conveys the precision of the estimate
- **Short confidence intervals** $\Longrightarrow$ **precise estimates**
- We have seen CIs for 2 parameters: mean (μ) and proportion (p)

Confidence Intervals

- You use a sample statistic to say sth about population parameter
- Sample statistic has a sampling distribution (different for each sample)

$$\hat{p} \sim N(p, \frac{p(1-p)}{n}) \qquad \bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

- But we do not know p and σ!
- We estimate them using the sample estimates $\hat{p}$ and s (from YOUR sample)

CIs: Formula

	Proportion	**Mean**
Population Parameter	p	μ
Sample Statistic	$\hat{p}$	$\bar{X}$
Standard error (SE)	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\frac{s}{\sqrt{n}}$
Critical Value	$z_{\alpha/2}$	$t_{\alpha/2,\ n-1}$
Confidence Interval (CI)	$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\bar{X} \pm t_{\alpha/2,\ n-1} \times \frac{s}{\sqrt{n}}$

CIs: Formula

When sample size conditions hold

- For proportion: $n\hat{p} > 10$ *and* $n(1-\hat{p}) > 10$
- For mean: $n > 30$ *or* $n > 10\ |K_4|$

$\Longrightarrow$ **You can use z critical value** so that formulas become

Variable Type	Statistic	CI Formula
Proportion/Binary	$\hat{p}$	$\hat{p} \pm z_{\alpha/2} * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Continuous	$\bar{X}$	$\bar{x} \pm z_{\alpha/2} * \frac{s}{\sqrt{n}}$

CIs: Interpretation

If I tell you: The $100(1-\alpha)\%$ confidence interval for the population parameter calculated from the sample is [a;b].

What does that mean?

$\Longrightarrow$ I'm $100(1-\alpha)\%$ **confident** that the value of the true population parameter is between [a;b].

$\Longrightarrow$ If we were to take 100 different samples and compute CIs for each sample, then approximately $100(1-\alpha)$ of the 100 confidence intervals will contain the true population parameter

* *Will only accept these two interpretations*

CIs: Interpretation

Other questions you may have about the interpretation

1. Could it happen that the population parameter is **outside** the interval [a;b]?
   - Yes, but this is NOT compatible with our sample at $100(1-\alpha)\%$ level of confidence
2. Could I be **more confident** ? Say from 95% to 99%?
   - Yes, but CIs will become wider, hence less precise estimate of the population parameter
   - [usually, not interesting and not done]

CIs: Interpretation

3. Could I be **more precise** at the *same level of confidence*?
   - Yes, a larger sample size will reduce the estimated standard error of the statistic, hence a narrower CI
   - [a larger sample is always better]

CIs: How To

1. You don't know the population parameter (μ)
2. You extract a sample and compute the sample statistic ($\bar{X}$)
3. You compute the standard error of the sample statistic ($se(\bar{X})$)
4. You are given the confidence level $100(1-\alpha)\%$
5. You apply the formula
6. You conclude by **copying** the interpretation we have seen together

Margin of Error

How large a sample do we need?

- I want to say something about the population parameter at 95% level of confidence.

▷ Extract a sample and compute a CI

- As before + I want to say sth within a given range (i.e. precision)

▷ Before CI, compute the needed sample size through the margin of error formula

Margin of Error

You don't care about t or z: you just have 2 as a value

- For sample mean: $ME = 2\frac{s}{\sqrt{n}}$
  - ▷ $n = \frac{4s^2}{ME^2}$
  - ▷ You need pilot sample to have s

- For proportion: $ME = 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
  - ▷ $n = \frac{4(1/2)^2}{ME^2} = \frac{1}{ME^2}$
  - ▷ You don't need pilot sample
  - ▷ 1/2 maximizes the standard error hence once we extract our sample, ME cannot be larger (it is equal or smaller)

Overview so far

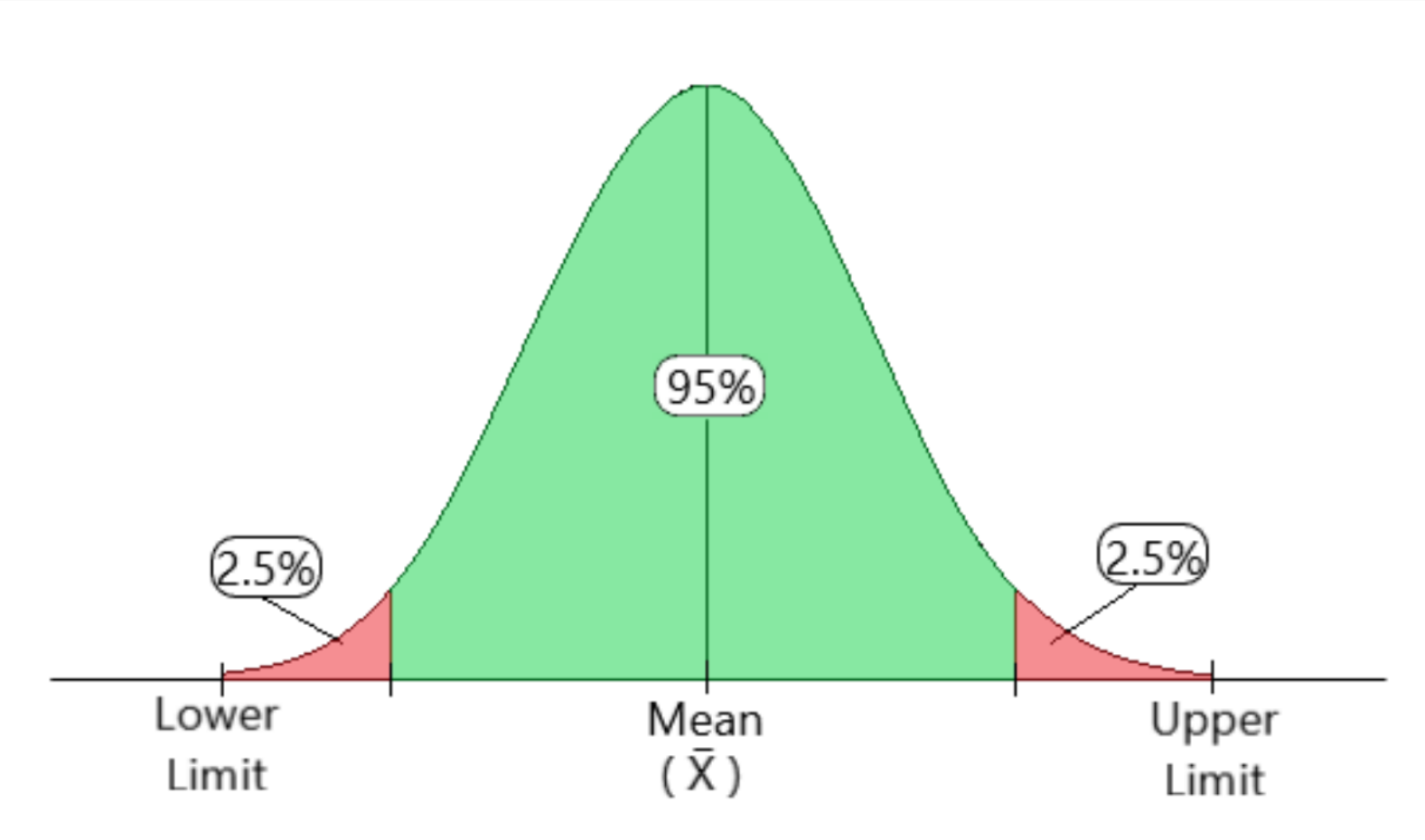
So far to take decisions: Control Limits, Confidence Intervals, and Margin of Error

Control Limits	Confidence Intervals	Margin of Error
We know the population parameters. We set control limits as a decision rule to determine when a sample statistic that we observe is "out of control," or suggests that the process of interest has likely changed.	We *do not* know the population parameters. We calculate a confidence interval to estimate a range of plausible values for a population parameter based on the sample.	We *do not* know the population parameters. We calculate a "back-of-the-envelope" 95% confidence interval by plugging in 2 instead of $z_{\alpha/2}$ or $t_{\alpha/2,n-1}$.

Excersise Break

CIs: Exercises

You should always have in mind the graph of what you are doing



Ch. 15: Exercise 40 (modified)

A city builds a new public parking garage in the central business district. It plans to pay for the structure through parking fees. During a 44-day period, the daily fees collected average \$1,264 with a standard deviation of \$150. Assume that the sample size conditions hold.

(a) Calculate a 95% CI for mean daily income from parking fees.

$\alpha = 0.05 \Leftrightarrow z_{\alpha/2} = 1.96$

$SE(\bar{X}) = \frac{s}{\sqrt{n}} = \frac{150}{\sqrt{44}}$

$$1,264 \pm 1.96 * \frac{150}{\sqrt{44}}$$

$$[1,219.7 \quad to \quad 1,308.3]$$

Ch. 15: Exercise 40 (modified)

A city builds a new public parking garage in the central business district. It plans to pay for the structure through parking fees. During a 44-day period, the daily fees collected average \$1,264 with a standard deviation of \$150. Assume that the sample size conditions hold.

(b) Would this interval be smaller or larger than a 90% CI? Do not actually calculate the 90% CI.

The 95% CI would be larger than a 90% CI. Indeed, the critical value for $\alpha = 0.10$ is $z_{\alpha/2} = 1.64$

(c) How could city planners achieve a narrower 95% CI?

Increase the sample size.

Ch. 15: Exercise 40 (modified)

(d) How many days' worth of data must planners collect to have 95% confidence of estimating the true mean plus \$10 or minus \$10? Does this seem like a reasonable objective?

From class we know, $ME = 2\frac{s}{\sqrt{n}}$ but don't learn this by heart!! Let's think about it.

The city planners must have a large enough sample so that the width of the CI is \$20. This condition implies that the difference between the upper bound and the mean (i.e., the length of one side of the CI) is \$10.

$$10 = 1.96 * \frac{150}{\sqrt{n}}$$

$$\sqrt{n} = 29.4$$

$$n = 864.36 \approx 865 days$$

This objective is not reasonable, as it would take almost three years to complete.

Review Ch. 16

Statistical Tests: Overview

- As usual: Want to say sth about population parameter but it is unknown and the best I can have is a sample

In a hypothesis test, we designate a **null hypothesis (H_0)** and an **alternative hypothesis (H_1)**

- H_0: preserving the status quo (i.e. taking no action)
- H_1: typically what we are looking to "prove" by "rejecting the null".

One-Sided vs. Two-Sided Tests

- One-sided tests have inequality: e.g., $H_0 : \mu \geq 5$
- Two-sided tests have equality: e.g., $H_0 : \mu = 5$

Statistical Tests: Overview

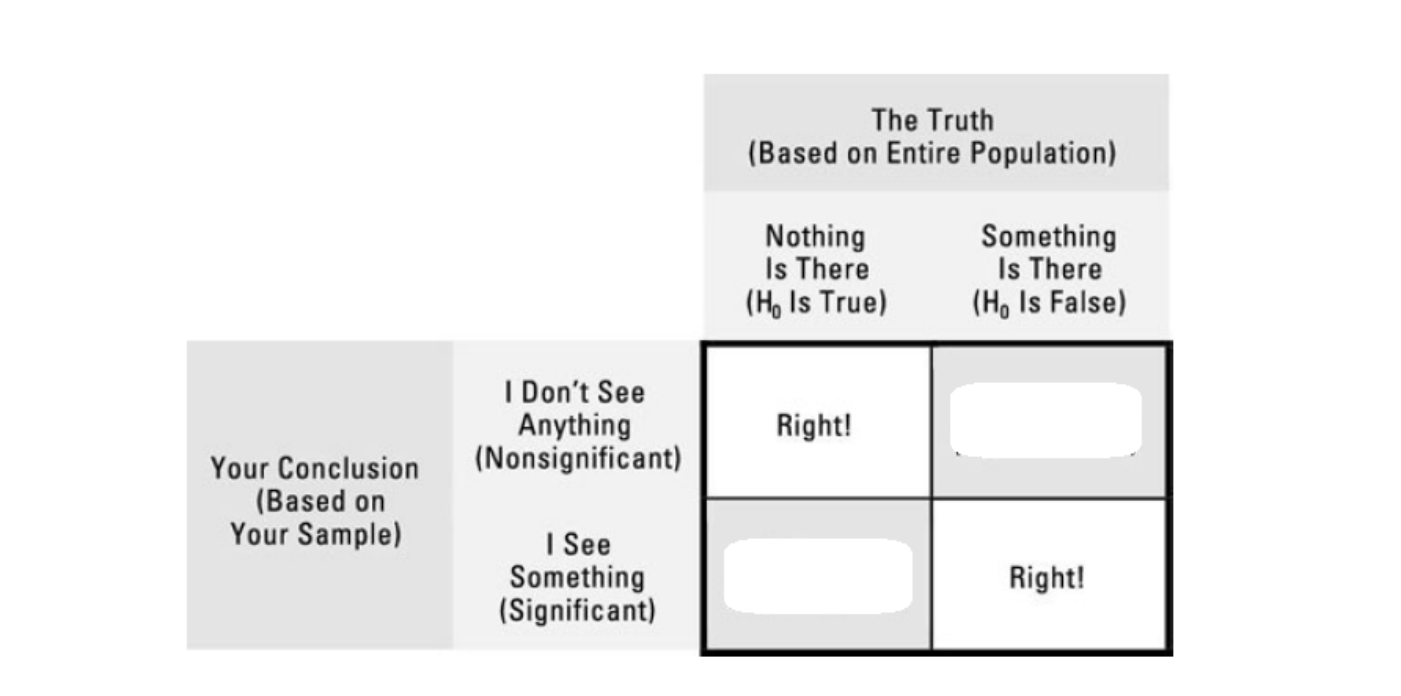
The conclusion of a hypothesis test is either:

1. we **reject** the null H_0
2. we **fail to reject** the null H_0

Note: If we reject H_0, for example with $\alpha = 0.05$, then we say that our result is statistically significant at the 5% level.

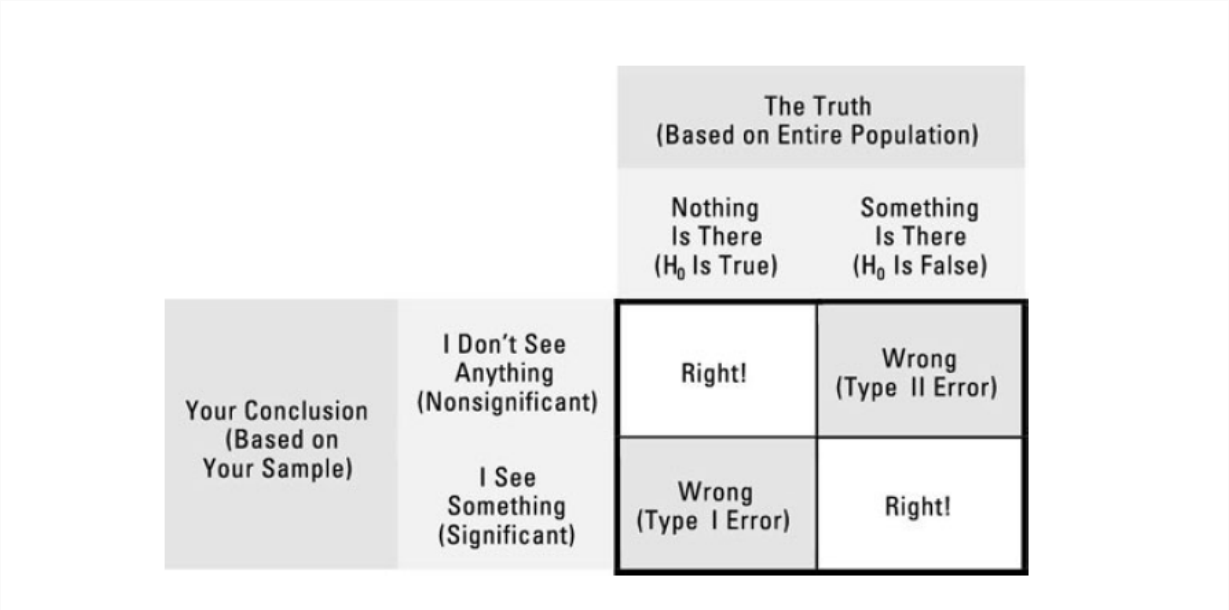
Our decision can be wrong. Thus, the concepts of Type I and Type II errors apply here as they did with control limits.

Statistical Tests: Overview



- If we reject the null, which was indeed true...
- If we do not reject the null but the null was indeed false...

Statistical Tests: Overview



		The Truth (Based on Entire Population)	
		Nothing Is There (H_0 Is True)	Something Is There (H_0 Is False)
Your Conclusion (Based on Your Sample)	I Don't See Anything (Nonsignificant)	Right!	Wrong (Type II Error)
	I See Something (Significant)	Wrong (Type I Error)	Right!

- If we reject the null, which was indeed true...

▷ **taking action when not needed** $\Longrightarrow$ **Type I**

- If we do not reject the null but the null was indeed false...

▷ **failing to take action when needed** $\Longrightarrow$ **Type II**

Warm-up Exercise

Spam affects productivity. The office manager is evaluating a commercial product that filters emails, reducing the amount of junk emails. The product is expensive, so the manager will only purchase it if it reduces the level of spam to less than 20%

$\Longrightarrow$ **What is the parameter of interest? What are the null and alternative hypotheses?**

- Parameter of interest: proportion of emails passing through the filter that are spam

Warm-up Exercise

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$\Longrightarrow$ **What is the parameter of interest? What are the null and alternative hypotheses?**

- Parameter of interest: proportion of emails passing through the filter that are spam
- $H_0: \ p \geq 0.20$
- $H_1: \ p < 0.20$
- This is one-sided test

Warm-up Exercise

$\Longrightarrow$ **What are Type I and Type II error in this case?**

- Type I:
- Type II:

Warm-up Exercise

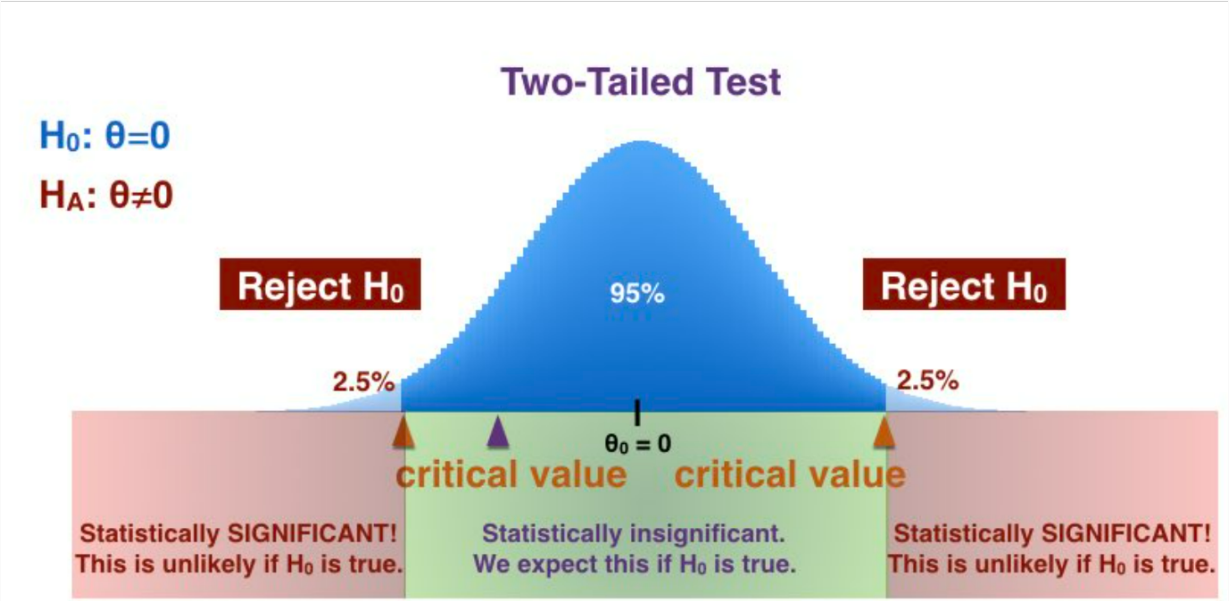
$\Longrightarrow$ **What are Type I and Type II error in this case?**

- Type I: Buying the license but not worth it/not cost effective
- Type II: Not buying the license which would have been cost effective (missed opportunity!)

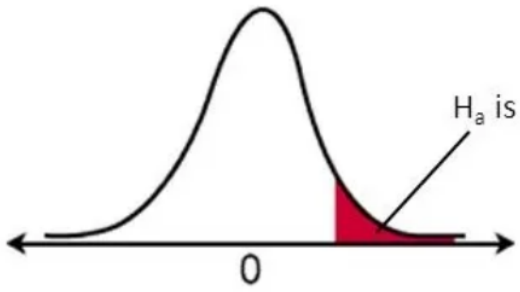
Statistical Tests: How to

- Statistical tests rely on the **sample** and corresponding **test statistic**

Key question: What is the chance of getting a **test statistic this far from H_0 if H_0 is true**?

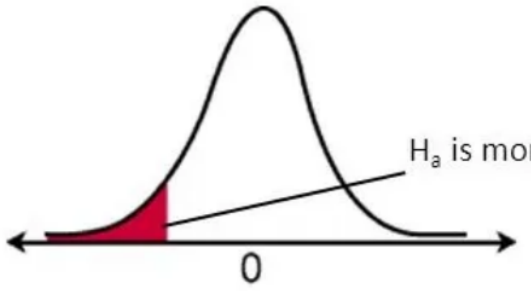


Statistical Tests: How to



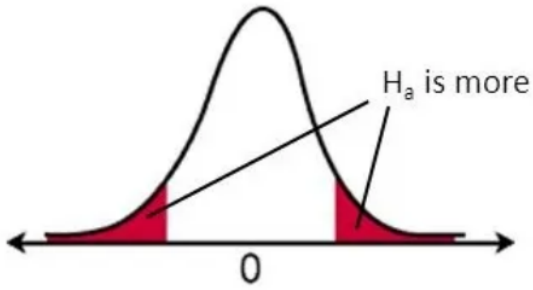
Right-tail test

$H_a: \mu > \text{value}$



Left-tail test

$H_a: \mu < \text{value}$



Two-tail test

$H_a: \mu \neq \text{value}$

Statistical Tests: How to

Implementation:

1. Check S.R.S. and Sample Size Condition
2. Specify H_0 and H_1
3. Set α - the maximum tolerance for Type I error
4. Calculate the test statistic (z or t-stat)
5. At this point, you have two options

Statistical Tests: How to

Option A (critical value)

6. Use Z-table or t-table to determine the **critical value for** α
   - One-sided test: Determine z_α or $t_{\alpha,\ n-1}$
   - Two-sided test: Determine $z_{\alpha/2}$ or $t_{\alpha/2,\ n-1}$
7. Compare **test statistic** (Step 4) with **critical value** (step 6) and apply following rejection rule...

Option B (p-value)

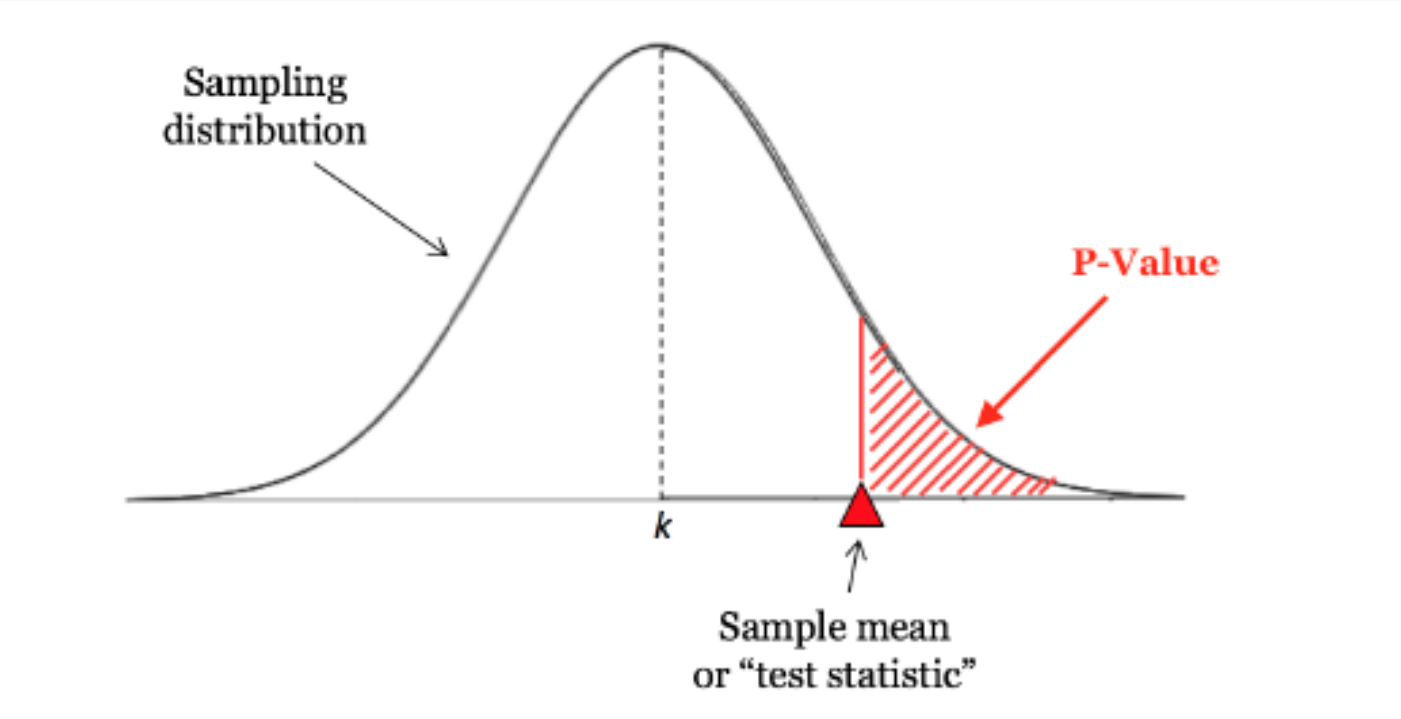
6. Use Z-table or t-table to determine the **p-value**

▷ *The probability of observing a statistic as extreme as the observed statistic given the null hp is true*

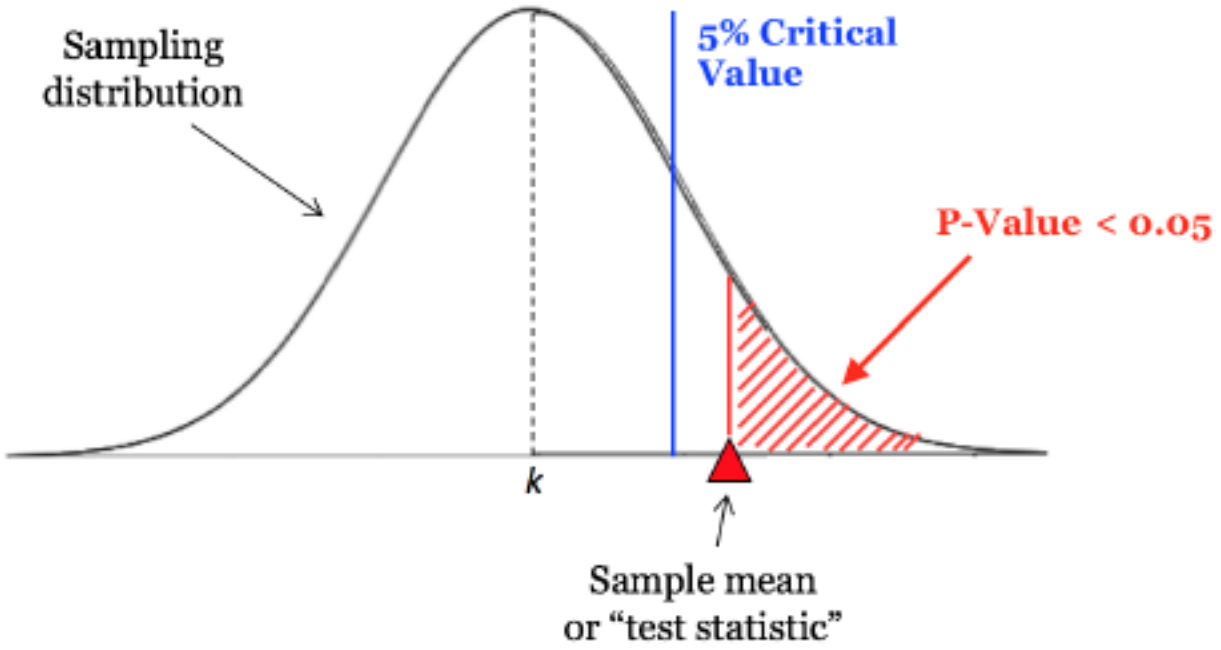
- P(Z ? your z-stat)
- ?: follows the inequality of H_1

7. Reject H_0 if:
   - p-value $< \alpha$ (one-sided test)
   - p-value $< \alpha/2$ (two-sided test)

Statistical Tests: How to

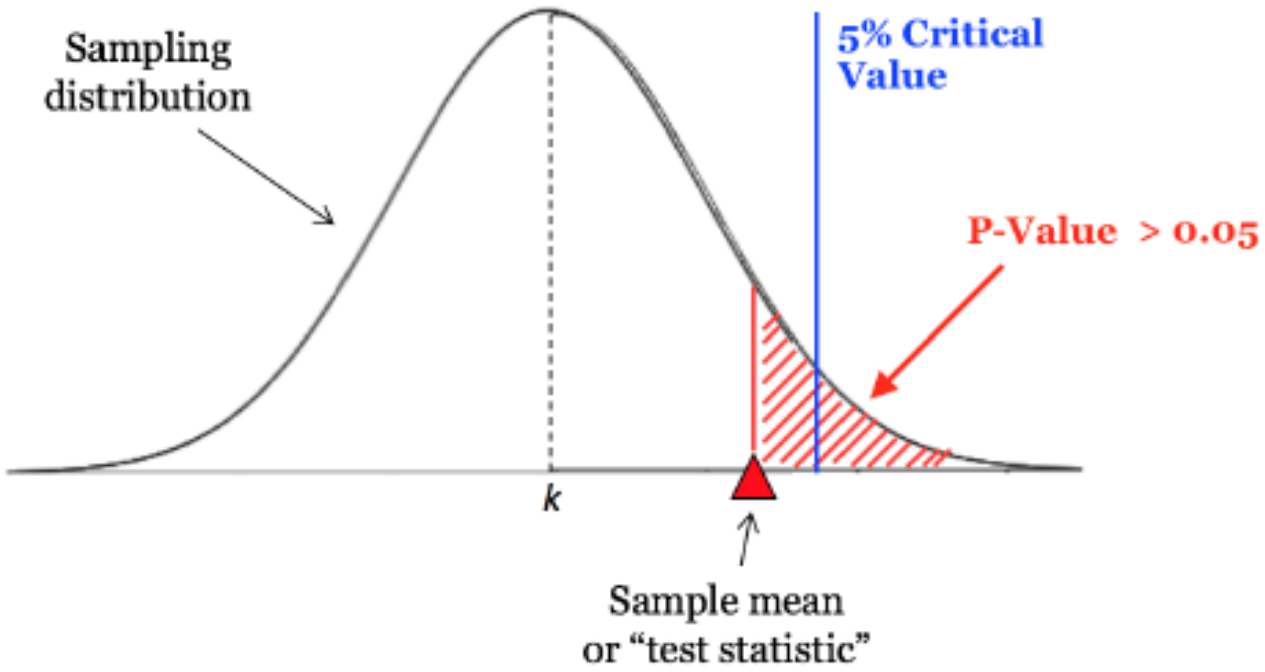


Statistical Tests: How to



When the $p - value$ is less than α, that means the test statistic is outside the decision criteria, so we reject the null

Statistical Tests: How to



When the $p - value$ is greater than α, that means the test statistic is inside the decision criteria, so we fail to reject the null

Statistical Tests: How to

Option B (critical value) - cont.

If **one-sided test**:

- H_1: population parameter $>$ value
  Reject H_0 if test statistic $>$ critical value
- H_1: population parameter $<$ value
  Reject H_0 if test statistic $<$ critical value

If **two-sided test**:

- H_1: population parameter $\neq$ value
  Reject H_0 if $|\text{test statistic}| >$ critical value

Statistical Tests: How to

Let's go back to point (4): calculate the test statistic

- **z-stat or t-stat?** it always depends on whether sample size conditions hold
- **How to?** Intuition is that you assume the statistic under the null H_0
- See next slide

Statistical Tests: How to

	Proportion	**Mean**
Population parameter	p	μ
Sample statistic	$\widehat{p}$	$\overline{X}$
Estimated standard error (se) of sample stat	$\sqrt{\frac{p_0(1-p_0)}{n}}$ Note: p_0 is the "cutoff" value	$s/\sqrt{n}$
Test statistic (z-stat or t-stat)	$z = \frac{\widehat{p}-p_0}{se(\widehat{p})}$ Note: $se(\widehat{p}) = \sqrt{\frac{p_0(1-p_0)}{n}}$; p_0 is the "cutoff" value	$t = \frac{\overline{X}-\mu_0}{se(\overline{X})}$ Note: μ_0 is the "cutoff" value
Conditions required to allow use of the Z-Table	(1) SRS (2) SSC: $n * p_0 > 10$, $n * (1 - p_0) > 10$	(1) SRS (2) SSC: $n > 10 \cdot \lvert K_4 \rvert$

CIs vs Hp Tests

Confidence intervals make positive statement about the population

- let the reader decide which interesting values

Hp tests make negative statement about the population

- rejecting H_0 tells us what μ is not

Excersise Break

Exercise: Costco

Costco is the third largest retailer in the United States. It spends hundreds of millions annually on electricity. Costco started last year retrofitting all of its stores' wall freezers to save energy. It is a large capital expense but one that Costco hopes will cut its energy costs substantially. So far, freezers have been replaced at a random sample of 100 stores.

Costco executives disagree on the cost-effectiveness of the retrofitting program. To pass a break even test, the freezers must reduce annual store energy expenditures by more than \$150,000. The sample mean is \$163,000 with a sample standard deviation (s) of \$52,000. Should Costco replace freezers at all of its stores?

(a) Is Costco interested in implementing a one-sided or two-sided test?

(b) State the null and alternative hypotheses.

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(a) Is Costco interested in implementing a one-sided or two-sided test?

Costco is interested in implementing a **one-sided** test.

(b) State the null and alternative hypotheses.

The parameter of interest is μ= average reduction in annual store energy expenditure

$$H_0 : \mu \leq 150,000$$

$$H_a : \mu > 150,000$$

Costco: Wholesale retailer

(c) Can we reject the null hypothesis at a 5% significance level?

Follow the steps laid out in the hypothesis test implementation slides above. (Try it yourself using Option B, too! I use Option A.)

1. We are told that Costco takes a S.R.S. And in this class, if not given the Kurtosis, then we can assume that the Sample Size Condition holds (and state it clearly in our response!).
2. Specify H_0 and H_a: See part (b)
3. Set α: We are told to use $\alpha = 0.05$
4. Calculate the test statistic (z-stat in our case): First, we calculate the standard error of the sample mean. Then, we calculate the z-statistic:

$$SE(\bar{X}) = \frac{52,000}{\sqrt{100}} = 5,200$$

$$z = \frac{163,000 - 150,000}{5,200} = 2.5$$

5. (Option A): Use the Z-table to determine the p-value for $z = 2.5$. We find $P(Z > 2.5) = 0.00621$.
6. Since $0.00621 < 0.05 = \alpha$, we reject the null hypothesis. Costco should invest in it.

Costco: Wholesale retailer

(d) Costco managers want to define a rule so that it can be applied to whatever sample they can extract. They want a rule such that once you compute the sample mean, you know whether you reject the null hypothesis at 5%. With a sample size of 100, what is this decision rule?

We determine the sample mean cutoff value ("action threshold") under the null hypothesis for rejecting the null hypothesis for $\alpha = 0.05$. For a one-sided test, $\alpha = 0.05$ corresponds to a critical value of $z = 1.645$.

$$\frac{C - 150,000}{5,200} = 1.645$$

$$C = 158,554$$

This action threshold means that Costco should act (i.e. invest in retrofitting) if the sample mean is greater than or equal to $158,554$. otherwise, they can preserve the status quo.

Costco: Wholesale retailer

(d) Suppose that the true population mean for the reduction in annual store expenditures from replacing the freezers is \$160,000. Costco has adopted a decision rule of the previous point with a sample size of 100. What is the probability that Costco would correctly reject the null hypothesis?

From the previous point, we know that Costco rejects the null (hence, investing in retrofitting) if the sample mean is greater than or equal to $158,554$. We want to computing the probability of correctly rejecting.

$$P(\bar{X} \geq 158,554 | \mu = 160,000)$$

$$P\left(\frac{\bar{X} - 160,000}{5,200} \geq \frac{158,554 - 160,000}{5,200}\right)$$

$$P(Z \geq -0.278) = 1 - P(Z < -0.278)$$

$$1 - 0.3905 = \mathbf{0.6095}$$

Note that the probability of correctly rejecting the null hypothesis (also known as the "power" of a test) is given by 1-P(Type II error).

$\Longrightarrow$ In this test, **P(Type II error)** = 0.3905

Review Ch. 17

Differences in Means or Proportions

- We typically evaluate the effect of a "treatment" using a difference in means (ATE): the difference in the average outcome for the treatment group and the average outcome for the control group.
- Everything we've done so far (test statistics, confidence intervals, control limits, etc.) applies in exactly the same manner.

Differences in Means or Proportions

But....What changes with two-sample tests?

- The statistic of interest: no longer just one statistic from a sample, now it is a difference between two statistics
- The comparison: now compare to a "break even difference", rather than a population mean or population proportion
- SE's: now remember to use the formula for SE(difference), rather than the simple formula for SE
- For means, degrees of freedom (t-tables!): the sum of the two sample sizes minus 2

Differences in Means: Formulas

	Difference in Means
Population Parameter	$\mu_1 - \mu_2$
Sample Statistic	$\bar{X}_1 - \bar{X}_2$
Sampling Distribution	$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$
Est. std. error (SE)	$se(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Null hypothesis	$H_0 : \mu_1 - \mu_2 \leq D_0$
Alt. hypothesis	$H_a : \mu_1 - \mu_2 > D_0$
Test Statistic	$t = (\bar{X}_1 - \bar{X}_2 - D_0)/se(\bar{X}_1 - \bar{X}_2)$
Reject H_0 if	p-value $< \alpha$ or $z > z_\alpha$
Sample Size Conditions	SRS and sample size conditions hold for both samples ($10\|K_4\| < n$ in both samples). To be interpreted causally, no lurking variable.
Example	Compare the average spending of customers offered a "prime" shipping promotional policy to the average spending of customers offered a standard shipping policy.

Differences in Proportions: Formulas

	Difference in Proportions
Population Parameter	$p_1 - p_2$
Sample Statistic	$\hat{p}_1 - \hat{p}_2$
Sampling Distribution	$\hat{p}_1 - \hat{p}_2 \sim \mathcal{N}\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$
Est. std. error (SE)	$se(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2}{n_2}}$
Null hypothesis	$H_0 : p_1 - p_2 \leq D_0$
Alt. hypothesis	$H_a : p_1 - p_2 > D_0$
Test Statistic	$z = (\hat{p}_1 - \hat{p}_2 - D_0)/se(\hat{p}_1 - \hat{p}_2)$
Reject H_0 if	p-value $< \alpha$ or $z > z_\alpha$
Sample Size Conditions	SRS and sample size conditions hold for both samples: $n_1\hat{p}_1 \geq 10$, $n_1(1-\hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, and $n_2(1-\hat{p}_2) \geq 10$
Example	Compare the proportion of customers who buy a product with an additional feature to the proportion of customers who buy a product when offered the standard features.

Questions?

Thank you!

